Multilevel Spatial-Channel Feature Fusion Network for Urban Village Classification by Fusing Satellite and Streetview Images

Runyu Fan, Jun Li, *Fellow, IEEE*, Fengpeng Li, Wei Han, and Lizhe Wang, *Fellow, IEEE*

***Abstract*— Urban villages (UVs) refer to areas of urban informal settlements lagging behind the rapid urbanization process. Recent studies focus on using satellite images to classify UV. However, satellite images only capture objects from a bird-eye perspective, thus cannot obtain complex spatial relationships between objects. In UV areas, buildings and objects are usually dense, small in size, and obscure each other. Therefore, it is challenging to classify UV accurately using only satellite images with bird-eye perspectives. In this article, to solve this problem, we proposed a novel method that uses satellite images combined with streetview images to classify UV. Specifically, we propose a novel multilevel spatial-channel feature fusion network, namely FusionMixer, that integrates CNN-based feature extraction modules and a multilevel spatial-channel feature fusing layer to make an optimal UV classification. Experiments were conducted in Shenzhen City (the RsSt-ShenzhenUV dataset) and a public UV dataset (the S^2UV dataset). The proposed FusionMixer achieved an increase of overall accuracy (OA) by 8.83% and 8.84%, and improves Kappa by 0.1765 and 0.1770 in the validation set and testing set, compared to the second-best fusion models in RsSt-ShenzhenUV dataset. Experiments in the S^2UV dataset show that the proposed FusionMixer improves OA by 1.82% and Kappa by 0.04 compared to other methods. We also added a set of experiments on a public dataset (Houston dataset) and compare our method with the current state-of-the-art multimodal fusion methods to prove the generalization of the proposed FusionMixer in fusing other multimodality data. These experiments confirmed the superior performance of the proposed FusionMixer.**

***Index Terms*— Deep learning, multimodal, remote sensing, streetview, urban village (UV).**

I. Introduction

ACCORDING to the definition of the United Nations Human Settlements Program, Urban villages (UVs) refer to high-density population urban informal settlements characterized by low standards and poverty. Urban areas carry the main human activities and social functions. Currently, more than half of the world's population lives in cities. Urbanization has brought about various social issues, including urban traffic conflicts, housing shortages, employment less, etc., which have caused urban development imbalances such as the widening gap between the rich and the poor. As a result, some urban populations live in UV with dense dilapidated houses, blocked traffic, unsafe drinking water, lack of public facilities, and poor medical conditions. UV Identification, which automatically recognizes the UVs and informal settlements in the city and gives the corresponding labels, is an important scientific problem that plays a significant role in the city's sustainable development, management, the estimation of the poor population, and future planning.

At the earliest, the identification of UV was done by the urban planning department using on-site investigation by staff. On-site investigation relies on great manual costs and material resources; thus, it is not timely and macroscopic. With the broad application of machine learning technology, scientists have proposed many well-performed methods and achieved remarkable results in various urban computing tasks, including urban land use and land cover [1], [2], [3], urban informal settlements clssification [4], [5], [6], urban functional zones mapping [7], [8], [9], population mapping [10], [11], [12], etc. Inspired by these progresses, some studies extend these methods to UV mapping tasks. These methods can be classified into machine learning-based (ML-based) and deep learning-based (DL-based) models.

UV usually carry some typical physical characteristics, such as small roof size, less floor, high density, limited space between shelter units, narrow roads, little vegetation, and other open spaces, etc. These typical physical characteristics provide a basis for models to identify UVs automatically. The ML-based methods were first applied to the UV mapping tasks and achieved excellent results. Initially, studies applied image interpreter-based methods to identify UV, such as object-based image analysis (OBIA) [13], [14], [15], [16], morphological profiles [17], [18], [19], gray-level-cooccurrence matrix (GLCM) [20], [21], [22], and achieved good performance. Apart from the image interpreter-based methods, a recent trend shows an increase in supervised learning-based machine learning methods, where researchers used support vector machine methods (SVMs) [23], [24], random forest (RF) models [25], [26], decision tree (DT) methods [25], [27], etc. ML-based

Manuscript received 12 June 2022; revised 14 August 2022; accepted 17 September 2022. Date of publication 20 September 2022; date of current version 3 October 2022. This work was supported in part by the National Natural Science Foundation of China under Grant 41925007 and Grant U21A2013, in part by the Special Fund of Hubei Luojia Laboratory under Grant 220100035, and in part by the Hubei Natural Science Foundation of China under Grant 2019CFA023. *(Corresponding author: Lizhe Wang.)*

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Digital Object Identifier 10.1109/TGRS.2022.3208166

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models are data-driven approaches that allow for repetitive learning from an extensive and rich training dataset with well-labeled samples.

Most ML-based methods work well because of human-designed representations and features. In this case, ML-based methods are good at optimizing weights to make an optimal prediction. But manually designed features are often incomplete, over-specified, and cost a lot of manual to design and validate. Meanwhile, ML-based models usually are not sufficiently accurate, which may be due to the inability to fit the complex nonlinear equations of the model.

DL-based methods are developed from neural networks, which exploit feature representation exclusively learned from input data. Recent advances in the field have proven deep learning a superior set of tools, sometimes even surpassing the human ability to solve highly computational tasks. These methods train a robust nonlinear deep neural network to learn and transform the feature from input data to fit the output and then use the well-trained model to analyze and make predictions. Studies have shown that CNNs [28], [29], [30], [31], [32], [33], VGG [34], [35], FCNs [36], [37], [38], UNet [5], [39], [40], and other DL-based models have achieved better results than previous methods in the UV area recognition task. However, these DL-based methods still have room for improvement when applied to UV identification task.

In real-world UV mapping scenarios, urban scenes under a high-resolution perspective are fragmented. In UV areas, buildings are dense, and houses are tiny and packed together, with complex spatial relations. It is difficult to identify UV using the single modality of remote sensing images. Given the insufficient accuracy of UV classification using single-modal satellite data, recent works started to study UV classification using multimodality data. For example, remote sensing and streetview images are studied to classify fine-grained UVs using a hierarchical approach [41] and a multimodal fusion dual-branch deep neural network [42]. Apart from streetview images, time-series human activity data have also been studied to promote UV classification with a hybrid Transformer-based spatio-temporal fusion network [4]. However, these methods ignored the interaction and fusion of spatial and channel features at different levels (low-level and high-level in deep neural networks) in the multimodality data fusion process. In this study, we found that a proper and simple fusion strategy of these multilevel spatial and channel features can greatly improve the performance of UV classification without using lots of complex designs in the above-mentioned models.

In this article, to solve the above problems, we proposed a novel multilevel spatial-channel feature fusion network, namely FusionMixer, that integrates a feature extraction module and a multilevel spatial-channel feature fusing layer to use satellite images combined with streetview images to recognize UV. Experiments were conducted in Shenzhen City (RsSt-ShenzhenUV dataset) and a widely used public UV dataset (the S^2UV dataset). We also added a set of experiments on a public dataset (Houston dataset) to show the generalization of the proposed FusionMixer in fusing other multimodality data. These experiments confirmed the superior performance of the proposed FusionMixer compared to the state-of-the-art models. The main contributions of this article are summarized as follows.

1) Focusing on the UV classification task, we have proposed a novel multilevel spatial-channel feature fusion network, namely FusionMixer, to use satellite images and streetview images to classify UV.
2) Different from previous models that based on CNN and Transformers, the proposed FusionMixer adopts mixer-based architecture for multimodality data fusing and obtains the best performance in three datasets.
3) The proposed FusionMixer enhanced UV classification by merging multilevel spatial-channel features learned from satellite and streetview images with a multilevel spatial-channel feature fusing layer.
4) The proposed FusionMixer improves classification accuracy on the public Houston dataset by fusing the multilevel spatial-channel features from hyperspectral and LiDAR data.

II. Related Work

UV classification, which aims to automate classify UVs in urban areas, is a significant scientific problem. It can be categorized into two classes according to the key models adopted, i.e., ML-based methods and DL-based methods; both are reviewed here.

A. ML-Based Methods for UV Classification

The ML-based methods formulate the UV classification task into a mathematical model and classify UV by solving this mathematical model. ML-based models regularly start with predefined or expert-built features and rules and then use machine learning models to train a classifier to identify UV.

Among the ML-based models, Taubenböck and Kraff [43] deliver an overview of the structural and morphological analysis of slums and state that building density, building sizes, and heights are significantly in slums. UV and slums have attributes of large building density, limited building size, and height. Kuffer *et al.* [15] conducted a survey of recent trends in UV mapping and indicates an increase in ML-based approaches.

In the early stage, image interpreter-based methods were studied to identify UV. Among them, OBIA models [13], [14], [15], [16] show effective performance due to the consideration of spectral, spatial, and geometric information of objects. By analyzing the morphological and texture characteristics of informal residential areas, a decent accuracy has been obtained in models using morphological profiles [17], [18], [19], and GLCM [20], [21], [22].

Apart from the image interpreter-based methods, supervised learning-based machine learning methods are increasingly applied in UV classification tasks. Researchers used powerful classifiers, such as SVMs [23], [24], RF [25], [26], DT [25], [27], etc., to identify informal settlements and achieved better results than previous image interpreter-based models. ML-based models allow repetitive learning in data-driven manners that learn from an extensive and rich training dataset

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with well-labeled samples to achieve competitive results in real-world UV mapping tasks.

However, these ML-based models' nonlinearity is not sufficient to effectively learn high-level representation from the labeled samples, these ML-based models are usually not sufficiently accurate. Besides, manually designed predefined or expert-built features are time-consuming and manual-cost. Thus, the above reasons limit the efficiency and accuracy when applying these methods to UV mapping tasks in large areas.

B. DL-Based Methods for UV Mapping

Due to the powerful nonlinear fitting ability and the capability to learn and transform features, DL-based methods have gained notable successes in various remote sensing tasks in recent years. DL-based methods can automatically learn the spectral and spatial information and characteristics of remote sensing and geospatial objects. These characteristics can be transformed to high-level representations to improve the models' performance compared with ML-based methods, without any manually designed features.

DL-based methods are based on deep neural networks, which have a powerful representation ability to learn from well-labeled input samples. Recent advances have proven the supervised DL-based models a superior set of algorithms, sometimes even outperforming human skills in various highly computational tasks. The supervised DL-based models train a robust deep neural network in an end-to-end manner to learn and transform an adaptive high-level feature representation and map the input data to the output labels.

Among these DL-based methods, studies consider the UV classification as the image classification task and propose a series of methods based on CNN. These CNN-based models utilize the variances of, such as CNNs [28], [29], [30], [31], [32], VGG [34], [35], etc., to identify UV from high-resolution remote sensing images. These CNN-based models can accurately distinguish UV by automatically learning the spatial information through CNN.

There are researchers study UV mapping from the perspective of semantic segmentation, using pixel-based semantic segmentation models, such as FCNs [36], [37], [38] and UNet [5], [39], [40]. These models can correctly identify the pixels corresponding to the UV areas from the high-resolution remote sensing images. Compared with the image classification-based models, the semantic segmentation-based models have more fine-scaled mapping results. However, since these semantic segmentation-based models are more complicate and time-consuming, applying them to UV classification tasks in large areas is challenging.

With the advance of remote sensing and geospatial technology, various data of different modalities can be easily obtained, and these data can reflect the different characteristics of objects from multiview perspectives. Studies begin to focus on UV classification using multimodal data. It is essential to design a proper model and integrating multimodalities remote sensing and geospatial data to extract UV areas. For example, a hierarchical approach has been proposed to fuse remote sensing and streetview images to classify fine-grained UV [41]. Besides, a multimodal fusion dual-branch deep neural network has also been proposed for UV classification based on satellite and streetview images [42]. Apart from static streetview images, dynamic time-series human activity data has also been studied to boost UV classification. Fan *et al.* [4] proposed a hybrid Transformer-based spatio-temporal fusion network to classify urban informal settlements using very-high-resolution (VHR) remote sensing images and time-series Tencent population density data. However, these methods ignored the interaction and fusion of spatial and channel features in the low-level (shallow) and high-level (deep) layers of deep neural networks during the multimodal data fusion process.

III. Proposed Method

To tackle the UV identification task, this article proposed a novel multilevel spatial-channel feature fusion network (FusionMixer), that integrates several CNN-based feature extraction modules and a multilevel spatial-channel feature fusing layer. This method aimed to fuse satellite images and streetview images to recognize UV.

The proposed FusionMixer is shown in Fig. 1. It consists of several CNN-based branches (the streetview images branch and the satellite images branch) to separately model the features from satellite images and streetview images, together with a multilevel spatial and channel fusion module based on Mixer architecture to learn and fuse the above different modality features to make optimal UV classification.

The streetview images branches and the satellite images branch utilize streetview images and satellite images, which are high-resolution and contain complex and detailed objects information. Former studies have indicated that CNNs can efficiently learn high-resolution spatial features due to the design of the shared convolution kernel and shifting invariant characteristics. Among these CNNs, the ResNet [44] models are widely adopted as a spatial feature extractor in image classification tasks owing to it is powerful residual learning ability and efficient design. Therefore, in this article, we use the ResNet model to obtain a preliminary spatial feature map of the streetview images and satellite images in the streetview images branches and the satellite images branch.

The features maps from the above branches are then fed into a multilevel spatial-channel fusion module to get an optimal UV classification. The CNN-based models have a common problem: when the depth reaches a certain amount, the features obtained from the deep layers tend to be global. Even if more convolutional layers are stacked, few diverse features can be received. Unlike the current mainstream data fusion model based on CNN, the multilevel spatial-channel feature fusion module provides a novel architecture of Mixer-based fusing. It adopts patch embeddings that are similar to ViT-transformer [45], the Swin-transformer [46] and MLP-Mixer [47] to learn more diverse features that are from shallow layers of CNNs. Then multiple Mixer layers [47] with skip connections are used to fuse the deep and diverse features to make an optimal classification.

The detailed architecture of the proposed FusionMixer is introduced in Section III-A and B: Spatial feature extraction via ResNet and feature fusing via multilevel spatial-channel feature fusion module.

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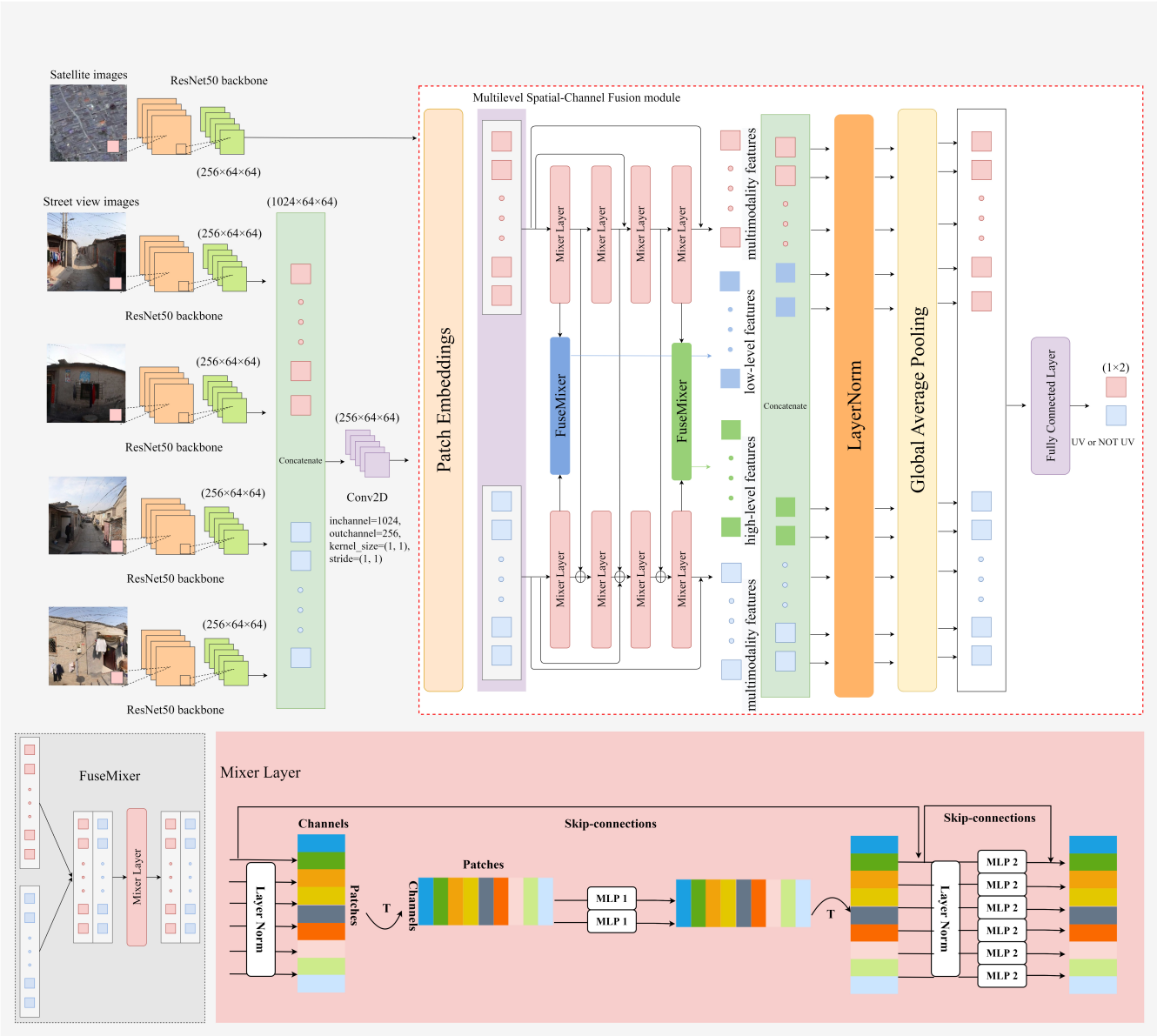


Fig. 1. Architecture of the proposed FusionMixer.

A. Spatial Feature Extraction via ResNet

High-resolution spatial features, including the details of land cover, ground objects, and buildings, are proven to affect UV mapping positively. The streetview images branch and the satellite images branch take streetview images, satellite images as input, respectively. The input size is unified into $(3 \times 256 \times 256)$, which means 256 pixels width, 256 pixels length, and three bands. These data are individually input into several identical ResNet50 network.

Considering the features obtained from the deep layers of ResNet50 tend to be global and less diverse, in the spatial feature extraction step, we only select the output feature map of the first residual module among the four residual modules of ResNet50, with the shape of $(256 \times 64 \times 64)$.

B. Multilevel Spatial-Channel Feature Fusion

Through the spatial feature extraction via ResNet step, several feature maps with the shape of $(256 \times 64 \times 64)$ from different data sources are obtained. These feature maps reflect the individual characteristics of different data sources to UV classification. To fuse the individual characteristics and model the combined influence, we proposed the multilevel spatial-channel feature fusion module to fuse the feature maps from satellite images and streetview images.

The multilevel spatial-channel feature fusion module module is shown in Fig. 1. It takes the feature maps from the streetview images branches and the satellite images branch as input. All the feature maps are with the shape of $(256 \times 64 \times 64)$. For the feature maps from the streetview images branches, we first concatenated these feature maps to a merged feature map with the shape of $(1024 \times 64 \times 64)$ and then passed it to a two-dimension (2-D) convolutional layer with the in_channels of 1024, out_channels of 256, kernel size of $(1, 1)$ and stride of $(1, 1)$. After that, the feature maps from these streetview image branches are merged into a feature map with the same shape, that is $(256 \times 64 \times 64)$, as the feature

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map from satellite images. The above two feature maps are then passed through two patch embedding layers to embed the feature maps into nonoverlapping patches with a shape of $(p \times h)$, where p and h refer to the number of patches and embedding dimension.

The patch embedding layers contains a 2D convolutional layer with the in_channels of 256, out_channels of h, kernel size of $(64/\sqrt{p})$ and stride of $(64/\sqrt{p})$. In this article, p and h is set to 16 and 256. After the above two patch embedding layers, two groups of embedded patches of satellite images features and streetview image features are obtained. Then a multilevel spatial-channel feature fusion module with Mixer layers and skip connections are added to fuse the embedded patches.

The illustration of the Mixer layer is shown in Fig. 1. The Mixer layer is a basic conponent of the MLP-Mixer [47]. It contains one token-mixing MLP (MLP 1 in Fig 1) and one channel-mixing MLP (MLP 2 in Fig 1), each consisting of two fully connected layers and a GELU [48] nonlinearity. The channel-mixing MLPs take individual rows of the feature as inputs independently on each token, to allow communication between different channels. The token-mixing MLPs take individual columns of the feature as inputs independently on each channel, to allow communication between different spatial locations (tokens). The Mixer layer can effectively and fully fuse the spatial-channel features by combining token-mixing MLP and channel-mixing MLP.

Mixer layers can be written as follows:

$$U_{*,i} = X_{*,i} + W_2\big(\mathrm{GLUE}\big(W_1\mathrm{LayerNorm}(X)_{*,i}\big)\big) \quad i \in \{1,\ldots,C\} \tag{1}$$

$$Y_{j,*} = U_{j,*} + W_4\big(\mathrm{GLUE}\big(W_3\mathrm{LayerNorm}(U)_{j,*}\big)\big) \quad j \in \{1,\ldots,S\} \tag{2}$$

where X denotes the input feature maps, $X \in \mathbb{R}^{S\times C}$. S is the number of patches and C is the number of channels. W denotes a MLP layer. U indicates the output of token-mixing MLP and Y indicates the output of channel-mixing MLP.

In Mixer layers, the tokens' hidden dimension is set to 128, and channels hidden dimension is set to 1024. In the proposed multilevel spatial-channel feature fusion module, two groups of four stacked Mixer layer are used to fuse the feature maps from satellite images and streetview images. For each group of stacked Mixer layer, we add a skip connection from the original feature map to the second and the last Mixer layer for more efficient feature communication between multilevels. And we added skip connections between the same levels of Mixer layers in different groups for efficient feature fusing between the feature maps from satellite images and streetview images.

To fuse the features from the above two different modalities, we feed the features from the first MixerLayer and the last MixerLayer of different modalities into a FuseMixer module and obtain two feature maps [each with a shape of $(p \times 2h)$] that integrate various modalities features at different levels. The feature maps from the first FuseMixer module are demoted as low-level features. The feature maps from the last FuseMixer module are demoted as high-level features.

TABLE I
DATA USED THE RSST-SHENZHENUV DATASET

Data Type	Time	Spatial Resolution	Data source
Satellite images	2020	$1.19m$	Google Earth
Streetview images	-	-	Baidu Map, Tencent Map

The FuseMixer module is shown in Fig. 1. FuseMixer first concatenates the feature map of different modalities and then feeds them into a MixerLayer to obtain the fused feature map.

Then, the output features from the above two stacked Mixer layer groups and two fused feature maps from FuseMixer are then concatenate to a merged feature map with a shape of $(p \times 6h)$ and fed into a layernorm layer. Then a Global Average Pooling [49] is added to each token and output a class specific vector with the shape of $(1 \times 6h)$. Finally, we added a fully connected layer with softmax activation to obtain the category vector with a shape $[1, 2]$. The value of each position in the category vector represents the probability value of the corresponding category, which means UV or not UV. We take the largest value in the category vector as the predicted category.

IV. EXPERIMENT AND ANALYSIS

To validate the effectiveness of the proposed FusionMixer, in this section, we have designed several sets of experiments on the RsSt-ShenzhenUV dataset, a public UV classification dataset (S^2UV dataset) and a public hyperspectral imagery (HSI)-LiDAR-based dataset (the Houston dataset). More information about the experimental data will be introduced in Section IV-A. Details about the experimental settings are presented in Section IV-B. The analysis and results in the RsSt-ShenzhenUV dataset are shown in Section IV-C. To demonstrate the generalization of the proposed method, we conduct experiments on a widely used public dataset, the S^2UV dataset. The experimental results in the S^2UV dataset are presented in Section IV-D. The experimental results in the Houston dataset are presented in Section IV-E.

A. Experimental Data

1) RsSt-ShenzhenUV Dataset: In this study, the RsSt-ShenzhenUV dataset is constructed and used to experiment and verify performance of the proposed model. In the process of building the RsSt-ShenzhenUV dataset, we first divide Shenzhen City into 6618 land use parcels according to the road network. Each land use parcel shares a category (UV or Not UV). Then the satellite images and streetview images, shown in Table I and Fig. 2, are spatially joined according to the land use parcels.

The satellite images are obtained from Google Earth with a spatial resolution of 1.19 m and acquired in 2020. The images are cropped by the land use parcels and then resized to image tiles of 256 × 256 pixels. In this way, the actual spatial resolution depends on the size of land use parcels. The distribution of the size of parcels is shown in Fig.3(a). In Fig. 3(a), each point means a sample with width and length

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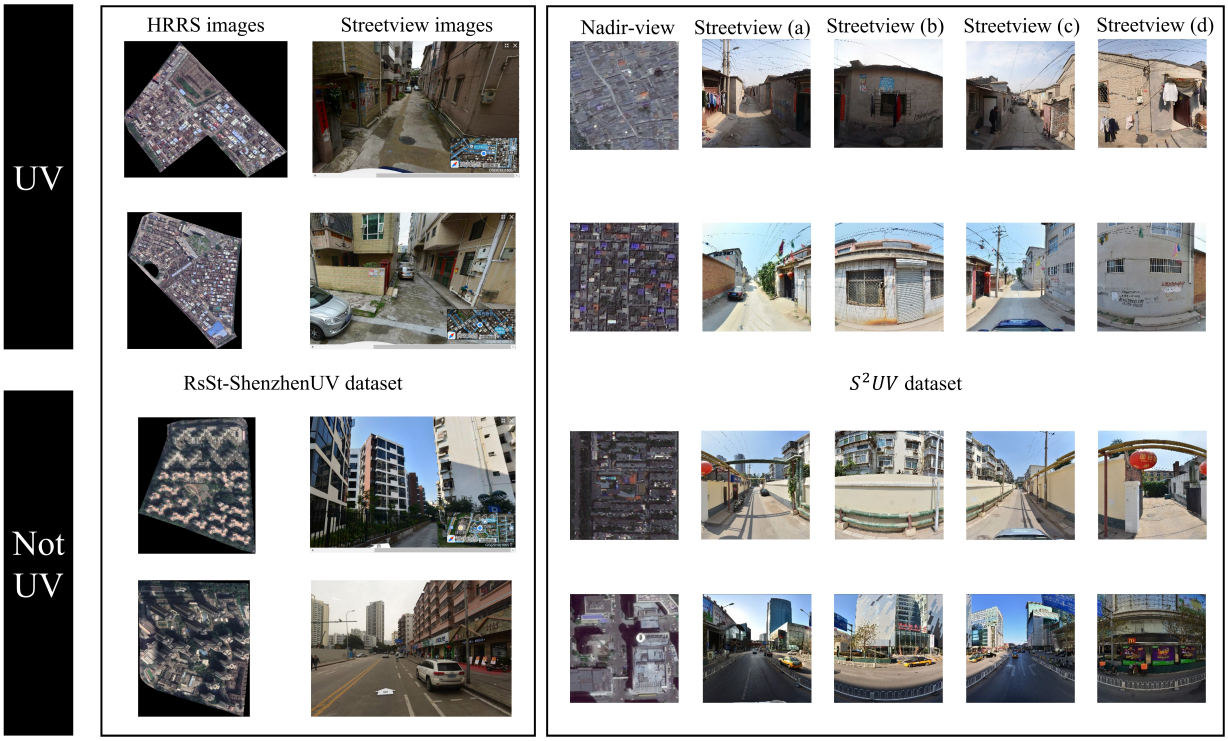
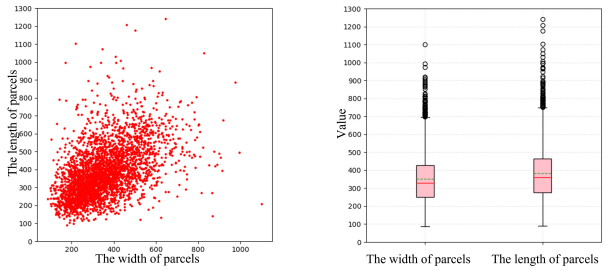


Fig. 2. Some samples in the RsSt-ShenzhenUV dataset and the S^2UV dataset.



(a) (b)

Fig. 3. Distribution and plotbox of the width and length of parcels. (a) Distribution of the size of parcels. (b) Plotbox of the size of parcels.

in the RsSt-ShenzhenUV dataset. The mean width value and mean length value of these parcels are about 350 and 383, the width value ranges from 87 to 1100, and the length value ranges from 91 to 1242. The RGB values of pixels outside the parcels are set to $[0, 0, 0]$.

In Fig. 3(b), the boxplots of the width and length of parcels are shown. Specifically, the box height in the boxplots indicates the interquartile range, and the upper and the bottom lines represent the maximum and minimum values of the width and length. The red line represents the median, and the green line represents the mean value. Black circles represent outlier samples. It can be observed from the box height range in the boxplots in Fig. 3(b) that the majority of the width values of parcels range from about 250 to 450, and the majority of the length values of parcels range from about 250 to 500.

The streetview images are obtained from Baidu streetview and Tencent streetview. By manually collecting the streetview images of each parcel, a total of 6618 streetview images were determined, and each streetview image corresponds to a parcel. The size of the acquired streetview images is inconsistent. In our method, the streetview images are resized to image tiles of 256 × 256 pixels to meet the input requirements of the model.

Through the preprocessing step above, 6618 parcels were obtained in total, with 6618 satellite images and 6618 streetview images. We then constructed the RsSt-ShenzhenUV dataset in a similar way as our previous work [4]. Specifically, we applied manual visual interpretation to determine whether each parcel is a UV and obtained 1618 UV samples. Considering the negative impact of sample imbalance, we randomly selected 1618 samples from the remaining 5000 samples that are Not UV, together with the 1618 UV samples to form the RsSt-ShenzhenUV dataset.

2) S^2UV Dataset: The S^2UV dataset dataset is a multi-modal dataset that was used for the UV classification task, which contains UV or Not UV samples in three core cities, which are short for Beijing City, Tianjin City, and Hebei Province in the Jing-Jin-Ji area. The dataset contains Google Earth level-17 high-resolution remote sensing imagery with a spatial resolution of 2.15 m and multiview Tencent streetview images from four field-of-view (i.e., 0°, 90°, 180°, and 270°), shown in Fig. 2.

The S^2UV dataset contains 856 and 1714 image samples corresponding to the two categories of UV and Not UV, respectively. In the S^2UV dataset, per sample contains four corresponding streetview images and one remote sensing image, both with a shape of 224 × 224 × 3. According to the previous work [42], the S^2UV dataset is divided into training

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TABLE II
RESULTS FOR DIFFERENT MODELS AND THE PROPOSED FUSIONMIXER

		Validation Set		Testing Set		
		OA (%)	Kappa	OA (%)	Kappa	Number of parameters
Individual models	MLP-Mixer with streetview images	73.06	0.4613	72.07	0.4414	18.3M
	MLP-Mixer with VHR remote sensing images	60.06	0.2012	56.69	0.1337	18.3M
	ResNet50 with streetview images	77.55	0.5511	75.51	0.5103	23.5M
	ResNet50 with VHR remote sensing images	81.26	0.6254	77.67	0.5535	23.5M
Fused models	two MLP-Mixer fusion	76.47	0.5394	73.71	0.4743	36.5M
	two ResNet50 fusion	78.17	0.5635	74.49	0.4897	47.0M
The proposed method	FusionMixer	**87.00**	**0.7400**	**83.33**	**0.6667**	40.3M

and testing sets using a split ratio of 7:3. Then the training and validation sets are separated from the training set using a split ratio of 8:2.

B. Experimental Settings

During the training, the Adam optimization method [50] was used, and the learning rate was set to 0.001. The batch size was set to 64, and a total of 100 epochs were trained. All implementations were based on Pytorch (Version torch == 1.8.2), torchvision (Version torchvision == 0.9.2), and Python (Version Python == 3.8.12) and an NVIDIA RTX 3090 GPU. To evaluate the proposed method, we evaluated the experimental results using the Kappa coefficient, the overall accuracy (OA), and the confusion matrix, which are broadly accepted evaluation criteria.

For the experiments in the RsSt-ShenzhenUV dataset, the following strategy was adopted to facilitate comparison to other models to split the dataset randomly: 20% as the training set (646), 20% as the validation set (646), and 60% as the testing set (1944). For the experiments in the S^2UV dataset, similar to the previous work [42], the S^2UV dataset is divided into training and testing sets using a split ratio of 7:3. Then the training and validation sets are separated from the training set using a split ratio of 8:2. Considering that the sample pairs in the RsSt-ShenzhenUV dataset only contains a satellite image and a single-view streetview image, therefore, in the experiment, the FusionMixer model used in the RsSt-ShenzhenUV dataset is different from Fig. 1, that is, the feature maps extracted by the streetview branch are directly fed to Patch Embeddings' layer without going through concatenate operator and Conv2D layer.

C. Experiments in RsSt-ShenzhenUV Dataset

Table II shows the OA of different models for UV classification in the testing set. The experimental results consist of two parts. The first parts are the individual models. These individual models are the modules that make up the proposed FusionMixer, including the streetview images branch, and the VHR remote sensing images branch, together with ResNet50 and MLP-Mixer model which are the basic component models of the proposed FusionMixer.

It can be seen that in these individual models, the ResNet50 with VHR remote sensing images had the highest OA and Kappa, with more than 21% improvement of OA and 0.42 of Kappa compared with the MLP-Mixer with VHR remote sensing images. Of all models mentioned, the OA and Kappa of the MLP-Mixer with VHR remote sensing images is the lowest, indicating that the only using MLP-Mixer model do not have competitive benefits for UV identification.

Both the two ResNet50 fusion model and two MLP-Mixer fusion model achieved an OA of more than 76%. However, because the two ResNet50 fusion model only uses a simple fusing strategy by concatenating the last layer of the two ResNet50 model to a joint classifier, the OA and Kappa are even lower than the ResNet50 with using only a single mode. It indicates that using a simple fusion strategy in this experiment does not necessarily increase OA and Kappa for UV identification.

The proposed FusionMixer improve OA by 8.83% and 8.84% in the validation set and testing set, and improves Kappa by 0.1770 and 0.1765 in the validation set and testing set, compared to the second-best fusion models. It shows that the proposed multilevel mixer fusion module in fusing VHR remote sensing images and streetview images greatly support distinguish UV.

The confusion matrices obtained by two MLP-Mixer fusion, two ResNet50 fusion, and the proposed FusionMixer are shown in Fig. 4(a)–(c), respectively. For all models compared, the informal settlements were heavily confused, especially in two ResNet50 fusion, with OA of 62.8%. In these compared models (two MLP-Mixer fusion and two ResNet50 fusion), the highest OA of UV is only 73.7%. It can be observed that the proposed FusionMixer had the highest OA in identifying informal settlements, with an OA improvement of 8.7% and 19.6% compared to two MLP-Mixer fusion model and two ResNet50 fusion model. This shows that the proposed multilevel mixer fusion strategy is efficient and accurate in fusing VHR remote sensing images and streetview images.

The results of UV mapping in the testing set is shown in Fig. 5. Fig. 5 shows a total of 6618 parcels in Shenzhen City, of which the testing set contains 1944 parcels. It can be seen

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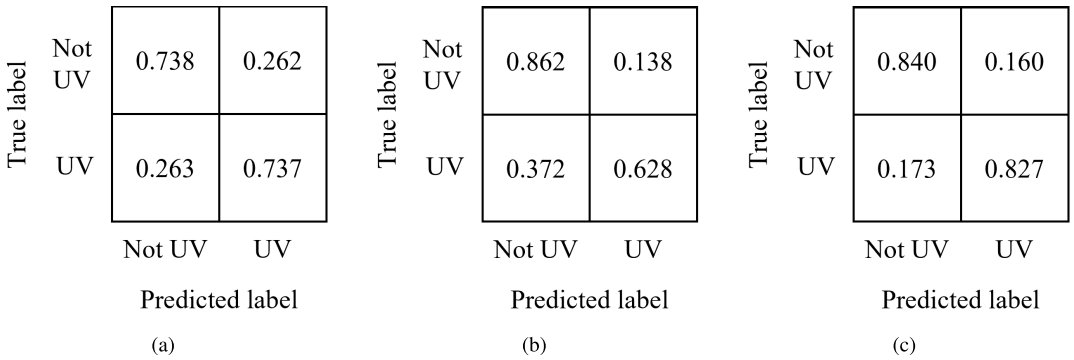


Fig. 4. Confusion matrix of the mentioned methods. (a) Confusion Matrix of the two MLP-Mixer fusion. (b) Confusion Matrix of the two ResNet50 fusion. (c) Confusion Matrix of the proposed FusionMixer.

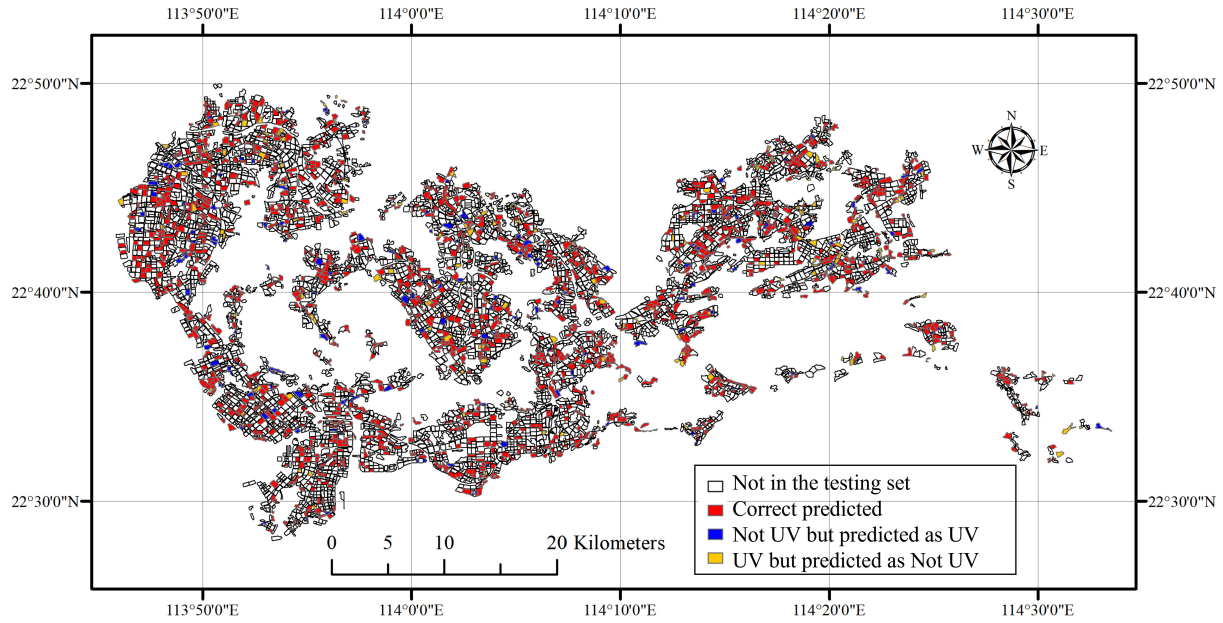


Fig. 5. Results of UV mapping in the testing set.

from the figure that 1620 parcels of the total 1944 parcels were correctly classified, and 156 parcels incorrectly classified Not UV as UV, and 168 parcels incorrectly classified UV as Not UV. It indicates that the proposed model has a quite stable classification performance in the entire Shenzhen City.

We further presented some examples of the classification results in Fig. 6. For parcels with VHR remote sensing image features that have highly distinguishable features of UV, such as A and B with density and tiny buildings in Fig. 6, models with VHR images outperformed models with streetview images. But for parcels with streetview features that have a large part of indistinguishable objects of UV, such as C and D with some self-built shanty buildings and narrow roads in Fig. 6, models with streetview images can avoid the misclassifications produced by the models with VHR remote sensing images. It can be seen that the misclassified examples using single modality data (VHR remote sensing images or streetview images) could be correctly classified after integrating the features by the FusionMixer.

D. Experiments in S^2UV Dataset

To prove the generalization of the proposed FusionMixer, we conduct experiments on a public UV classification dataset, the S^2UV dataset, and compare it with some recently proposed models, including models with single modalities and models with multiple modalities. Table III shows the OA and Kappa of different models in the testing set. It can be seen that, for both the single modality model using satellite images and the single modality model using streetview images, the ViT model [45] achieved the best performance, with an OA of 90.80% and Kappa of 0.7939 using satellite images and an OA of 90.03% and Kappa of 0.7602 using streetview images. Compared with Transformer-based ViT model, the CNN-based models (i.e., VGG-16 [51], ResNet-50 [44], and DenseNet-121 [52]) achieve less significant results. It indicates the superior performance of Transformer to CNN.

For multiple modality model, the Trans-MDCNN [42] get better performance with an OA of 92.61% and Kappa of 0.8352 compared with single modality model. The proposed FusionMixer model obtained the best OA of 94.43% and

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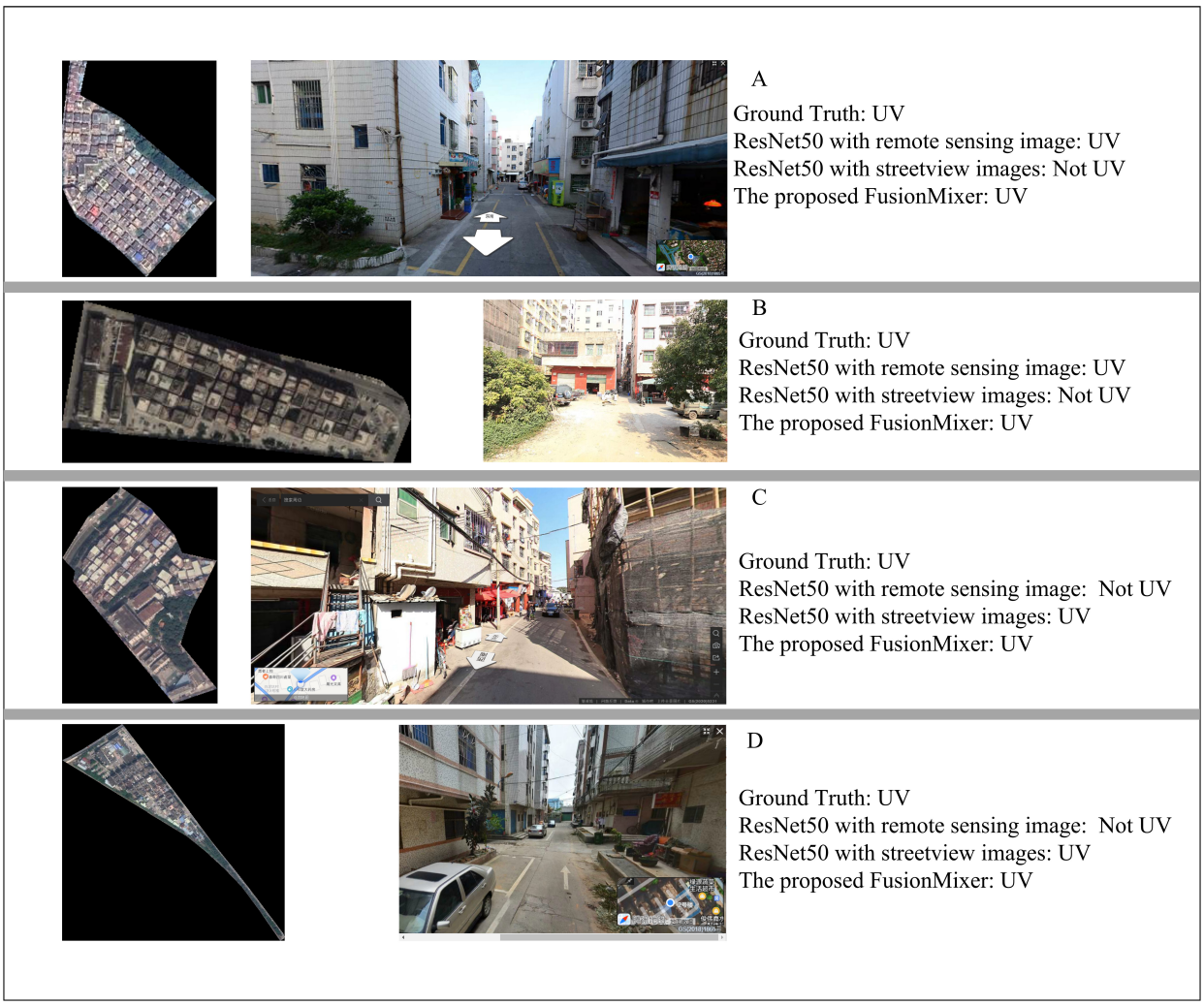


Fig. 6. Examples of the UV classification results.

TABLE III
RESULTS FOR DIFFERENT MODELS AND THE PROPOSED FUSIONMIXER

	Methods	OA (%)	Kappa
Models with satellite images	VGG-16	87.17	0.7029
	ResNet-50	89.63	0.7662
	DenseNet-121	88.99	0.7464
	ViT	90.80	0.7939
Models with streetview images	VGG-16	83.55	0.6126
	ResNet-50	85.10	0.6404
	DenseNet-121	86.65	0.7028
	ViT	90.03	0.7602
Fused models	Trans-MDCNN [42]	92.61	0.8352
	FusionMixer (ours)	**94.43**	**0.8757**

Kappa of 0.8757, improving OA of 1.82% and Kappa of 0.0405. These results indicate that the proposed FusionMixer can enhance the model's accuracy and obtain better classification results by fusing multilevel spatial-channel features of satellite images and streetview images.

TABLE IV
RESULTS FOR FUSING SINGLE-VIEW AND MULTIVIEW IMAGES WITH SATELLITE IMAGES

	Methods	OA (%)	Kappa
FusionMixer (ours)	Satellite & Single-view (0°)	92.23	0.8254
	Satellite & Single-view (90°)	91.84	0.8117
	Satellite & Single-view (180°)	93.65	0.8610
	Satellite & Single-view (270°)	92.49	0.8305
	Satellite & Multi-view	**94.43**	**0.8757**

Furthermore, we have discussed the impact of using satellite and single-view streetview images on UV classification. The streetview images consist of multiview images, which could capture a panoramic perspective of UV. To analyze the impacts of different views of streetview images in UV classification, we made a series of comparisons of fusing different views of streetview images and satellite images. The experiments results are as follows.

Table IV shows that the average OA and Kappa of four field-of-view (i.e., 0°, 90°, 180°, and 270°) fusing with satellite

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TABLE V
ABLATION STUDY OF MULTILEVEL FEATURES

Method	multimodality features	low-level features	high-level features	OA (%)	Kappa
FusionMixer	√			94.17	0.8704
FusionMixer	√	√		93.91	0.8606
FusionMixer	√	√	√	**94.43**	**0.8757**

images. When only fusing single-view of streetview images and satellite images, the combination of satellite images and single-view (270°) streetview images obtained the best performance with an OA of 93.65% and Kappa of 0.8610. If all the views of streetview images are involved, the OA would increase to 94.43% with an average improvement of about 2.6% to 0.8%, and the Kappa would increase to 0.8757 with an average improvement of about 0.064 to 0.0147. The reason is that using a single-view image could only capture partial information about UV, resulting in less significant performance in UV classification. If multiview images are introduced to get a complete appearance of UV with additional information, the accuracy can be improved. This verifies the effectiveness of multiview streetview images in UV classification.

Ablation study is used to evaluate the effect of various components in the proposed FusionMixer on the overall performance. The experimental results in the S^2UV dataset are shown in Table V. It can be observed that the FusionMixer with only multimodality features from streetview branches and satellite branch achieved an OA of 94.17% and a Kappa of 0.8704. When adding the low-level features from the shallow FuseMixer layer, a performance decrease in OA and Kappa of 0.26 and 0.0098 can be observed. It indicates fusing only multimodality features and low-level features would not improve UV classification performance. The performance had an improvement of about 0.52% in OA and 0.0151 in Kappa when adding the high-level features from the deep FuseMixer layer. This indicates the superior of fusing multimodality features, low-level features, and high-level features together in UV classification.

E. Experiments and Comparison on the Houston Dataset

To prove the generalization of the proposed FusionMixer in fusing other multimodality data, we further conduct experiments on a widely used public dataset, the Houston dataset. The Houston dataset is delivered by GRSS Data Fusion Contest 2013. This dataset was captured on the University of Houston campus by airborne sensors with HSI that consists of 144 spectral bands and one band LiDAR images. This dataset contains 15 029 ground-truth samples of 15 different landuse categories. The dataset is divided into a training set with 2832 pixels and a testing set with 12 197 pixels.

In the experiments, the input HSI patch and LiDAR patch in Houston dataset are set to the same size (11 × 11) with former studies for a fair comparison. The former-used ResNet50 backbone in Fig. 1 contains Conv2D with the kernel size of 7 × 7 and MaxPool2d layers with a kernel size of 3 × 3 and stride of 2. The above Conv2D and MaxPool2d layers both reduce the size of the feature map. As a result, when the input patch is with the size of (11 × 11), the ResNet50 backbone

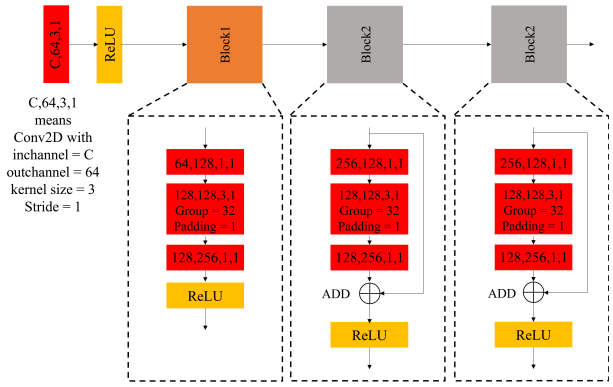


Fig. 7. ResNet backbone the proposed FusionMixer on the Houston dataset.

cannot work. Due to the difference in the input size, there are some differences with the version of FusionMixer in the UV classification experiment.

Specifically, in this experiment, we replaced the ResNet50 backbone (as shown in Fig. 1) with a ResNet backbone (as shown in Fig. 7) that removes Conv2D layers with the kernel size of 7 × 7 and MaxPool2d layers. The ResNet backbone uses the residual network architecture which is similar to the ResNet50 backbone above. The ResNet backbone first utilizes a Conv2D layer to reduce the patch size from 11 × 11 to 9 × 9, then employs three simple CNN modules which does not contain MaxPool2d layers. The convolution kernel size is set to 3 × 3 and 1 × 1 with padding operation to keep the feature map to the size of 9 × 9.

The ResNet backbone takes images of shape ($C \times 11 \times 11$) as input and outputs a feature map of ($256 \times 9 \times 9$). C is set to 144 in HSI images and 21 in LiDAR images (the attribute profiles [59] extracted from the LiDAR image, resulting in 21-band profiles). In this experiment, the HSI images play the role of satellite images, and the LiDAR images play the role of street view images in Fig. 1. The Mixer layer used in this experiment is the same as the one in the UV classification, except the patch size p is set to 3.

To prove the effectiveness of the proposed FusionMixer, we compare it with some recently proposed models, including models with single modalities and models with multiple modalities. Table VI shows the OA and Kappa of different models in the testing set. It can be seen that, for the single input model, the Mini-GCN model achieved the best OA of 86.02% and Kappa of 0.8399. For multiple input model, the MAHiDFNet get the best OA of 90.06% and Kappa of 0.8924. The proposed FusionMixer model obtained the best OA of 92.17% and Kappa of 0.9151, improving OA of 2.11% and Kappa of 0.0227. The proposed method achieved an accuracy of 85% in all categories except the Railway category. In the

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TABLE VI
RESULTS FOR DIFFERENT MODELS AND THE PROPOSED FUSIONMIXER ON THE HOUSTON DATASET

	Single Input Model			Multiple Input Model			The proposed method
Class	SVM [53]	CNN-PPF [54]	Mini-GCN [55]	S2FL [56]	MDL-RS [57]	MAHiDFNet [58]	FusionMixer
Health grass	82.43	85.57	82.81	81.67	83.10	83.10	**86.51**
Stressed grass	82.05	98.21	81.20	82.99	81.58	82.61	**99.15**
Synthetic grass	99.80	98.42	100.00	100.00	100.00	100.00	**100**
Trees	92.80	97.73	91.19	92.05	**99.72**	93.37	95.64
Soil	98.48	96.50	98.58	99.43	**99.81**	99.72	99.72
Water	95.10	97.20	89.51	99.30	95.10	**100.00**	95.80
Residential	75.47	85.82	77.43	76.31	90.02	**92.26**	90.02
Commercial	46.91	56.51	83.67	68.09	87.94	**91.55**	90.98
Road	77.53	71.20	79.98	69.22	81.59	86.02	**87.06**
Highway	60.04	57.12	56.27	52.12	86.68	66.31	**91.41**
Railway	81.02	80.55	**96.11**	90.42	89.37	95.92	81.88
Parking lot 1	85.49	62.82	92.41	87.42	85.69	**94.24**	91.26
Parking lot 2	75.09	63.86	68.42	70.18	83.16	**98.60**	92.63
Tennis court	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	92.71
Running track	98.31	98.10	98.52	98.31	98.73	**100.00**	**100**
OA%	80.49	83.33	86.02	81.95	89.60	90.06	**92.17**
Kappa × 100	78.98	81.88	83.99	80.40	88.75	89.24	**91.51**

Highway category, the proposed method is the only model that reached an OA upper than 90%. These results indicate that the proposed FusionMixer can enhance the model's accuracy and obtain better classification results by fusing multilevel features of HSI images and LiDAR data.

V. CONCLUSION

In this article, we proposed a novel method, namely FusionMixer, that uses satellite images and streetview images to jointly classify UV. Specifically, in the proposed FusionMixer, streetview images are used as an auxiliary and processed with satellite images separately by a feature extraction module and then fused via the multilevel spatial-channel feature fusing layer to make an optimal UV classification. To assess the performance of the proposed FusionMixer, several models were compared. Experiments were conducted in Shenzhen City (RsSt-ShenzhenUV dataset), a public UV dataset (S^2UV dataset) and a public HSI-LiDAR-based dataset (the Houston dataset).

The experiments confirmed the superior performance of the proposed FusionMixer by existing works. The experimental results confirmed a consistent conclusion that by fusing satellite images and streetview images using the proposed multilevel spatial-channel feature fusion module, the proposed FusionMixer had the best performance in both quantity and quality with an increase of OA by 8.83% and 8.84% in the validation set and testing set, and improves Kappa by 0.1765 and 0.1770 in the validation set and testing set, compared to the second-best fusion models in RsSt-ShenzhenUV dataset. Besides, we add a set of experiments on the S^2UV dataset and compare our method with the current state-of-the-art multimodal fusion methods. Experiments show that the proposed FusionMixer improves OA by 1.82% and Kappa by 0.04 compared to other methods. To prove the generalization of the proposed FusionMixer in fusing other multimodality data, we add a set of experiments on the Houston dataset and compare our method with the current state-of-the-art multimodal fusion methods. Experiments show that the proposed FusionMixer improves OA by 2.1 and Kappa by 0.023 compared to other methods. These experiments prove the superiority of the proposed FusionMixer.

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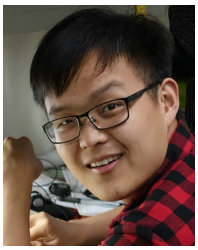
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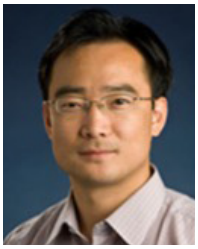
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